

given $f(r, s(r))$,

$$\frac{d}{dr} f = \frac{df}{dr} + \frac{df}{ds} \frac{ds}{dr}$$

$$E(r, s) = f_p(r) + f_s(r) \cdot \sum_n T_n(s)$$

$$\frac{dE}{dr_{ij}} = \frac{d}{dr} f_p + \frac{d}{dr} (f_s) \cdot \sum_n T_n(s) + f_s \sum_n \frac{dT_n(s)}{ds}$$

$$\frac{dT_n(s)}{dr} = \frac{dT_n(s)}{ds} \cdot \frac{ds}{dr} \quad \text{2-body only}$$

$$\Rightarrow \frac{dE_{ij}}{dr_{ij}} = \frac{d}{dr} f_p(r_{ij}) + \frac{d}{dr} f_s(r_{ij}) \sum_n T_n(s) + f_s \sum_n \frac{dT_n(s)}{ds} \frac{ds}{dr}$$

For more than two particles we need the partial derivatives wrt all relevant rods:

$$\frac{\partial E}{\partial r_{ij}}, \frac{\partial E}{\partial r_{ik}}, \frac{\partial E}{\partial r_{jk}} \rightarrow 3 \text{ body}$$

$$\frac{\partial E}{\partial r_{ij}}, \frac{\partial E}{\partial r_{ik}}, \frac{\partial E}{\partial r_{il}}, \frac{\partial E}{\partial r_{jk}}, \frac{\partial E}{\partial r_{jl}}, \frac{\partial E}{\partial r_{kl}} \rightarrow 4 \text{ body}$$

Take 3 body: $r_1 = r_{ij}$ $r_2 = r_{ik}$ $r_3 = r_{jk}$

$$\Rightarrow E_{ijk} = f_s(r_1) f_s(r_2) f_s(r_3) \sum_{\alpha} \sum_{\beta} \sum_{\gamma} \overset{E \rightarrow \text{symmetric}}{C_{\alpha\beta\gamma}} T_{\alpha}(s_1) T_{\beta}(s_2) T_{\gamma}(s_3)$$

$$\hookrightarrow E_{ijk}(r_1, r_2, r_3) \quad * \rightarrow \text{symmetrical}$$

$$\nabla E = \frac{\partial E}{\partial r_1} + \frac{\partial E}{\partial r_2} + \frac{\partial E}{\partial r_3} \quad \text{3-body only}$$

$$\frac{dE}{dr_1} = f_s(r_1) f_s(r_2) f_s(r_3) \sum_{\alpha} \sum_{\beta} \sum_{\gamma} \overset{E}{C_{\alpha\beta\gamma}} \frac{dT_{\alpha}(s_1)}{ds_1} \frac{ds_1}{dr_1} T_{\beta}(s_2) T_{\gamma}(s_3) + \frac{\partial}{\partial r_1} [f_s(r_1) f_s(r_2) f_s(r_3)] \sum_{\alpha} \sum_{\beta} \sum_{\gamma} C_{\alpha\beta\gamma} T_{\alpha}(s_1) T_{\beta}(s_2) T_{\gamma}(s_3)$$

this pattern is easily extensible for more bodies

But the second gradient is also needed:
2-body:

$$\frac{dE_{ij}}{dr_{ij}} = \frac{d}{dr} f_p(r_{ij}) + \frac{d}{dr} f_s(r_{ij}) \sum_n T_n(s) + f_s \sum_n \overset{u}{C_n} \frac{dT_n(s)}{ds} \overset{v}{\frac{ds}{dr}}$$

$$\frac{d^2 E_{ij}}{dr_{ij}^2} = \frac{d^2}{dr^2} f_p + \frac{d^2}{dr^2} f_s \sum_n T_n(s) + \frac{d}{dr} f_s \sum_n \overset{u}{C_n} \frac{dT_n(s)}{ds} \overset{v}{\frac{ds}{dr}} + \frac{d}{dr} f_s \sum_n \overset{u}{C_n} \frac{dT_n(s)}{ds} \overset{v}{\frac{ds}{dr}} + f_s \sum_n \left[\overset{u}{\frac{d^2}{ds^2} T_n(s)} \overset{v}{\left(\frac{ds}{dr} \right)^2} + \overset{u}{\frac{d}{ds} T_n(s)} \overset{v'}{\frac{d^2 s}{dr^2}} \right]$$

$$\frac{d^2 E_{ij}}{dr_{ij}^2} = \frac{d^2}{dr^2} f_p + \frac{d^2}{dr^2} f_s \sum_n C_n T_n(s) + 2 f_s \sum_n \overset{u}{C_n} \frac{dT_n(s)}{ds} \overset{v}{\frac{ds}{dr}} + f_s \sum_n \left[\overset{u}{\frac{d^2}{ds^2} T_n(s)} \left(\frac{ds}{dr} \right)^2 + \overset{v}{\frac{d}{ds} T_n(s)} \frac{d^2 s}{dr^2} \right] \quad \text{2 body second gradient}$$

$$\begin{aligned} \frac{dE}{dr_1} &= f_s(r_1) f_s(r_2) f_s(r_3) \sum_{\alpha} \sum_{\beta} \sum_{\gamma} \overset{E}{C_{\alpha\beta\gamma}} \frac{dT_{\alpha}(s_1)}{ds_1} \frac{ds_1}{dr_1} T_{\beta}(s_2) T_{\gamma}(s_3) + \frac{\partial}{\partial r_1} [f_s(r_1) f_s(r_2) f_s(r_3)] \sum_{\alpha} \sum_{\beta} \sum_{\gamma} C_{\alpha\beta\gamma} T_{\alpha}(s_1) T_{\beta}(s_2) T_{\gamma}(s_3) \\ \frac{d^2 E}{dr_1^2} &= \frac{\partial}{\partial r_1} \left[f_s(r_1) f_s(r_2) f_s(r_3) \sum_{\alpha\beta\gamma} C_{\alpha\beta\gamma} \frac{dT_{\alpha}(s_1)}{ds_1} \frac{ds_1}{dr_1} T_{\beta} T_{\gamma} \right] + f_s(r_1) f_s(r_2) f_s(r_3) \sum_{\alpha\beta\gamma} C_{\alpha\beta\gamma} \left[\frac{d^2}{ds_1^2} T_{\alpha}(s_1) \frac{ds_1}{dr_1} \frac{ds_1}{dr_1} + \frac{dT_{\alpha}(s_1)}{ds_1} \frac{d^2 s_1}{dr_1^2} \right] T_{\beta} T_{\gamma} \\ &\quad + \frac{\partial^2}{\partial r_1^2} [f_s(r_1) f_s(r_2) f_s(r_3)] \sum_{\alpha\beta\gamma} C_{\alpha\beta\gamma} T_{\alpha}(s_1) T_{\beta} T_{\gamma} + \frac{\partial}{\partial r_1} [f_s(r_1) f_s(r_2) f_s(r_3)] \sum_{\alpha\beta\gamma} C_{\alpha\beta\gamma} \frac{dT_{\alpha}(s_1)}{ds_1} \frac{ds_1}{dr_1} T_{\beta} T_{\gamma} \end{aligned}$$

$$\begin{aligned} \frac{d^2 E}{dr_1^2} &= f_s(r_1) f_s(r_2) f_s(r_3) \sum_{\alpha\beta\gamma} C_{\alpha\beta\gamma} \left[\frac{d^2}{ds_1^2} T_{\alpha}(s_1) \left(\frac{ds_1}{dr_1} \right)^2 + \frac{dT_{\alpha}(s_1)}{ds_1} \frac{d^2 s_1}{dr_1^2} \right] T_{\beta} T_{\gamma} \\ &\quad + 2 \frac{\partial}{\partial r_1} [f_s(r_1) f_s(r_2) f_s(r_3)] \sum_{\alpha\beta\gamma} C_{\alpha\beta\gamma} \frac{dT_{\alpha}(s_1)}{ds_1} \frac{ds_1}{dr_1} T_{\beta} T_{\gamma} \\ &\quad + \frac{\partial^2}{\partial r_1^2} [f_s(r_1) f_s(r_2) f_s(r_3)] \sum_{\alpha\beta\gamma} C_{\alpha\beta\gamma} T_{\alpha}(s_1) T_{\beta} T_{\gamma} \end{aligned}$$

Smoothing function derivatives:

$$f_s = \begin{cases} 0 & r > r_c \\ \frac{1}{2} + \frac{1}{2} \sin\left(\pi \frac{r-d}{r_c-d} + \frac{\pi}{2}\right) & \text{otherwise} \end{cases}$$

$$\theta(r) = \pi \frac{r-d}{r_c-d} + \frac{\pi}{2} \quad \frac{d\theta}{dr} = \frac{\pi}{r_c-d}$$

$$\frac{df_s}{dr} = \frac{1}{2} \cos \theta \frac{\pi}{r_c-d} = \frac{\pi}{2(r_c-d)} \cos\left(\pi \left(\frac{r-d}{r_c-d} + \frac{\pi}{2}\right)\right)$$

$$\frac{d^2 f_s}{dr^2} = -\frac{\pi^2}{2(r_c-d)^2} \sin\left(\pi \left(\frac{r-d}{r_c-d} + \frac{\pi}{2}\right)\right)$$

Penalty function derivatives

$$f_p = \begin{cases} A(r_c+d-r)^3 & r < r_c+d \\ 0 & \text{otherwise} \end{cases}$$

$$\frac{df_p}{dr} = -3A(r_c+d-r)^2$$

$$\frac{d^2 f_p}{dr^2} = 6A(r_c+d-r)$$

Note for both f_s & f_p derivatives are 0 for anything but the "otherwise" pieces.